

Module 5

Two-Port Networks and Network Graphs

Dr. Elizabeth Rufus

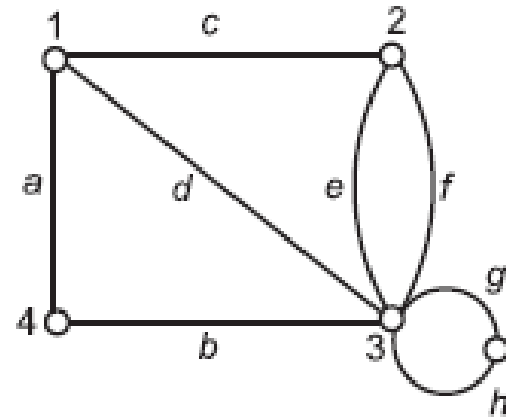
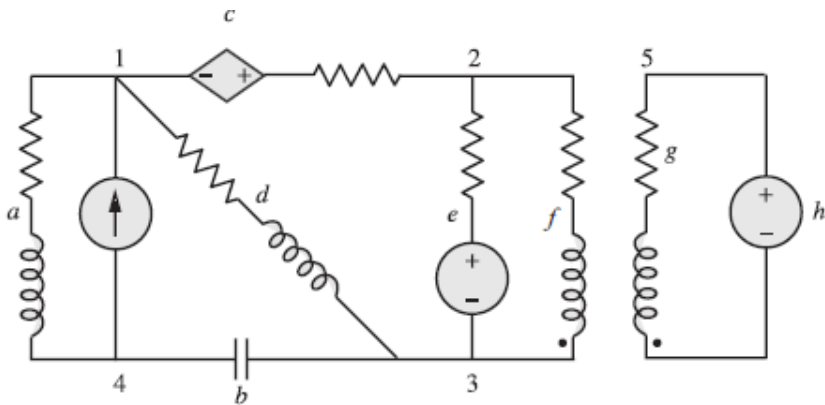
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Network Graphs - terminology

Network: A combination of two or more network elements is called a network.

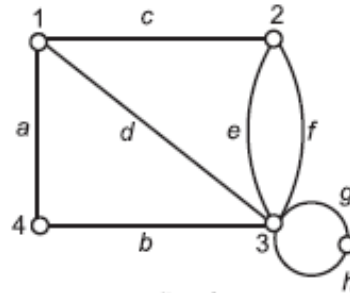
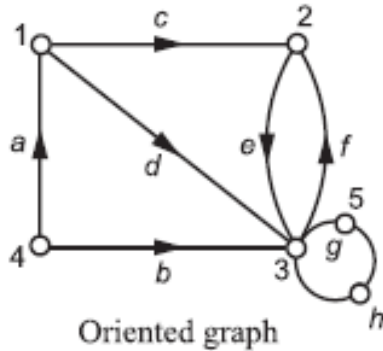
Circuit: If the network has at least one closed path, it is a circuit. Every circuit is a network but every network is not a circuit.

Graph: When all the elements in a network like Resistors, Inductors, Capacitors etc are replaced by line segments with their end points shown as dots or circles, voltage source with short circuit and current source with open circuit is called the graph of the network.



Network Graphs

Directed (or Oriented) graph: A graph is said to be directed (or oriented) when all the nodes and branches are numbered or direction assigned to the branches by arrow.



Node: A point at which two or more elements have a common connection.

Degree of Node: Number of branches incident on it.

Path: A set of elements that may be traversed in order without passing through the same node twice.

Branch: A single path, containing one simple element, or a combination of elements which connects one node to any other node. A simple line segment with it's two distinct end points (Nodes) represent a branch. It does not indicate anything about the nature of the element/s .

Loop: A closed path in the oriented graph is called as loop.

Mesh: A loop which does not contain any other loops within it.

Network Graphs

Tree: It is that part of a Graph with an interconnected open set of branches which include all the nodes of the given graph. In a tree of the graph there cannot be any closed path.

Properties of a Tree:

- (i) It consists of all the nodes of the graph.
- (ii) If the graph has N nodes, then the tree has $(N-1)$ branch.
- (iii) There will be no closed path in a tree
- (iv) There can be many possible different trees for a given graph depending on the no. of nodes and branches.

Cotree: Those branches that are not part of the tree form the cotree, or complement of the tree.

Twig (Tree branch): All branches of a tree are called Twigs. They are drawn using continuous lines.

Link (Cotree branch): All branches of the cotree are called the links. They are drawn using broken lines.

Network Graphs

Note that: Nodes are drawn as dots

Twigs are shown dark lines

Links are shown dotted lines

Relation between nodes, branches, twigs and links:

N = no. of nodes

L = total no. of links

B = total no. of branches

No. of twigs = $N-1$

Then, $L = B - (N-1)$ or $L = B - N + 1$

There are L branches in the cotree and $(N - 1)$ branches in the tree.

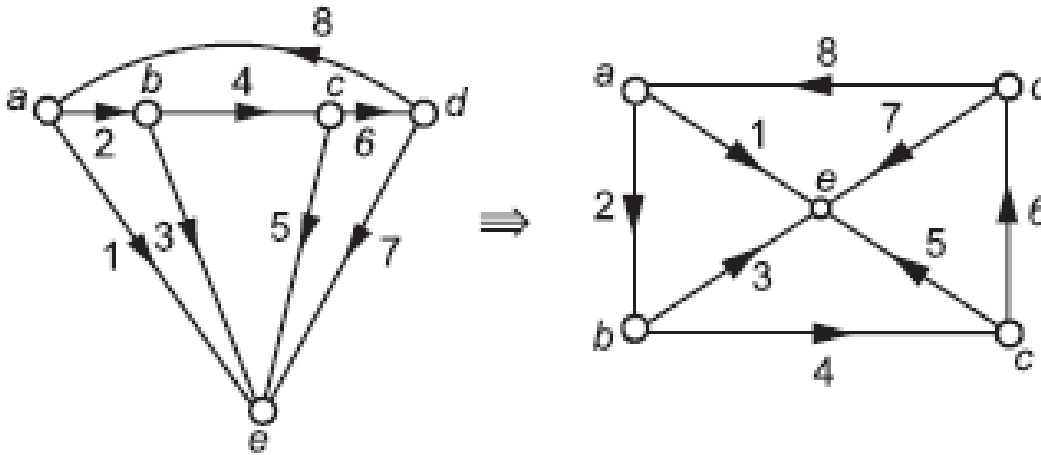
Network Graphs

Isomorphic graphs:

Two graphs are said to be isomorphic if they have the same incidence matrix, though they look different.

It means that they have the same numbers of nodes and the same numbers of branches.

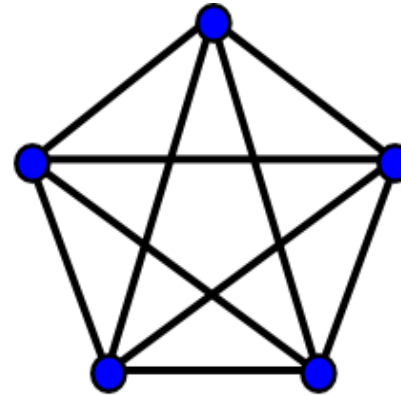
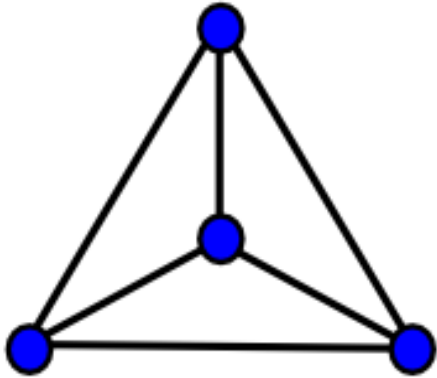
There is one to one correspondence between the nodes and one to one correspondence between the branches.



Network Graphs

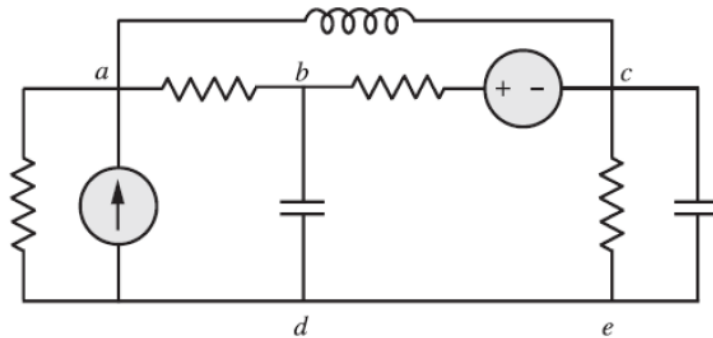
Planar circuit: A circuit which may be drawn on a plane surface in such a way that no branch passes above or beneath any other branch.

Non-planar circuit: Any circuit which is not planar.

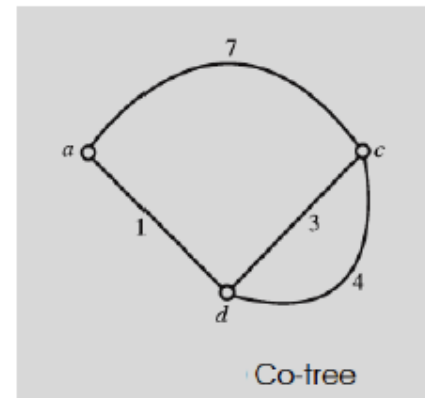
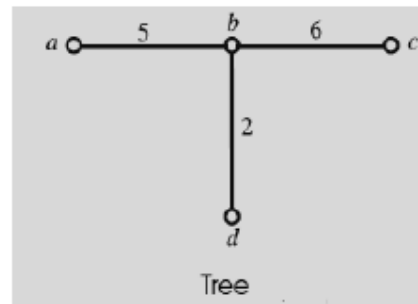
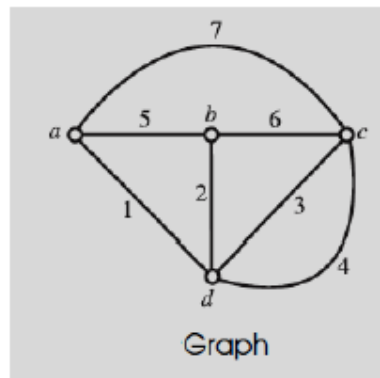


Network Graphs

1. Draw the graph, tree and the co-tree for the circuit given below.

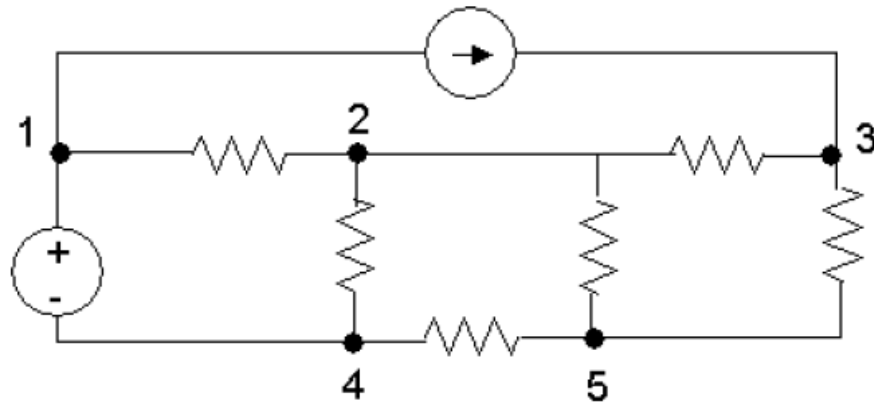


One tree and cotree is given below:

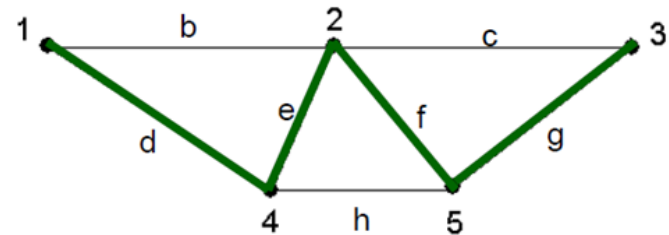
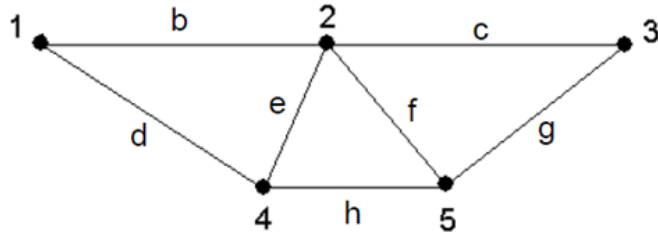


Network Graphs

2. Draw the graph, tree and its co-tree.



One tree (green line) and cotree is given below:



Incidence Matrix

Any oriented graph can be described completely in a compact matrix form.

Here we specify the orientation of each branch in the graph and the nodes at which this branch is incident. This branch is called incident matrix.

When one row is completely deleted from the matrix the remaining matrix is called a reduced incidence matrix.

Order of incidence matrix is $(N \times B)$.

Incidence matrix is also known as **augmented matrix**.

Properties of incidence matrix:

1. Number of non zero entries of row indicates degree of the node.
2. The non zero entries of the column represents branch connections.
3. If two columns has same entries then they are in parallel.

Procedure to form incidence matrix with elements a_{ij} :

$a_{ij} = 1$, if j th branch is incidence to i th node and direction is away from node.

$a_{ij} = -1$, if j th branch is incidence to i th node and direction is towards from node.

$a_{ij} = 0$, if j th branch is not incidence to i th node .

Reduced Incidence Matrix

Significance of the incidence matrix lies in the fact that it can translate all the geometrical features in the graph into an algebraic expression

When one row is eliminated from the complete incidence matrix, the remaining matrix is called reduced incidence matrix.

Any node of a connected graph can be selected as a reference node.

Then the voltages of the other nodes (referred to as buses) can be measured with respect to the assigned reference.

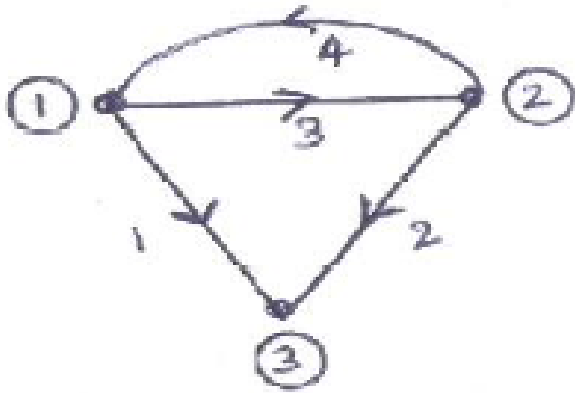
The matrix obtained from A by deleting the row corresponding to the reference node is the element Reduced incident matrix A_R and is a matrix with size of $(N-1) \times B$.

The number of possible trees in a network is given by $A_R A_R^T$.

where A_R is the reduced incidence matrix and A_R^T is the transpose of A_R

Incidence Matrix

1. Obtain the complete incidence matrix for the graph.

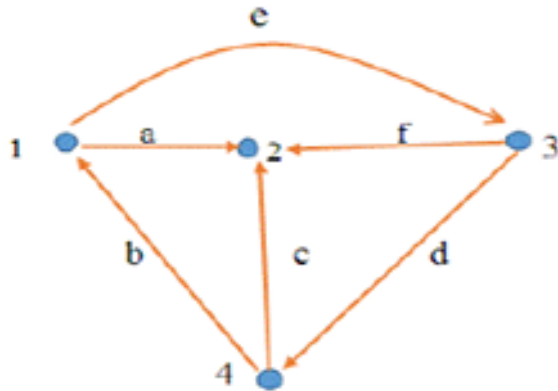


Nodes	Branches			
	1	2	3	4
1	1	0	1	-1
2	0	1	-1	1
3	-1	-1	0	0

$$A_n = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ -1 & -1 & 0 & 0 \end{bmatrix}$$

Network Graphs

2. Obtain the incidence matrix and the reduced incidence matrix for the given graph.



In this directed graph, there are 4 nodes and 6 branches. Thus the incidence matrix for the above graph will have 4 rows and 6 columns.

$[A] =$

nodes \ branches	a	b	c	d	e	f
1	1	-1	0	0	1	0
2	-1	0	-1	0	0	-1
3	0	0	0	1	-1	1
4	0	1	1	-1	0	0

Note that sum of all elements in each column is zero.

Network Graphs

Reduced Incidence Matrix: If from a given incidence matrix $[A]$, any arbitrary row is deleted, then the new matrix formed will be reduced incidence matrix.

It is represented by symbol $[A_R]$. The order of reduced incidence matrix is $(N-1) \times B$ where N is the number of nodes and B is the number of branches.

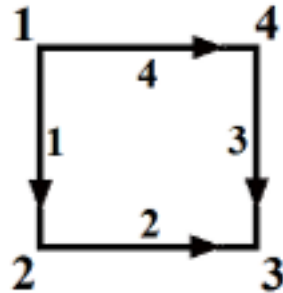
$[A_R] =$

nodes \ branches	a	b	c	d	e	f
1	1	-1	0	0	1	0
2	-1	0	-1	0	0	-1
3	0	0	0	1	-1	1

Note that row 4 is deleted from $[A]$.

Network Graphs

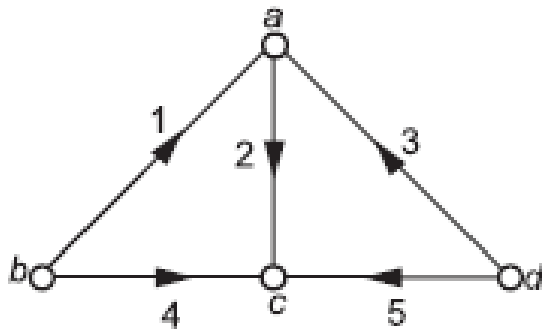
3. Obtain the incidence matrix for the given graph.



The incidence matrix A is written as:

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}.$$

4. Obtain the complete incidence matrix for the graph.



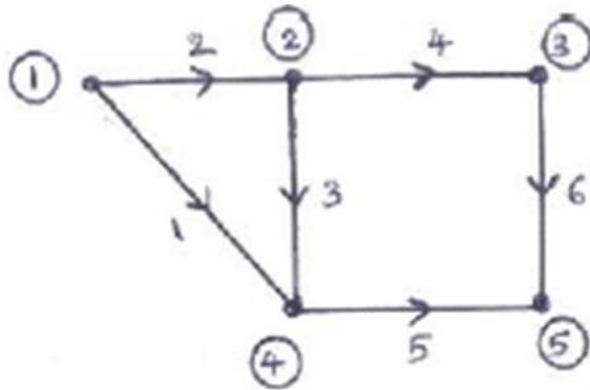
Nodes ↓

branches

$$A_n = \begin{array}{c} \begin{matrix} a \\ b \\ c \\ d \end{matrix} \end{array} \begin{array}{ccccc} \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{bmatrix} -1 & 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

Network Graphs

5. Write the complete and reduced incidence matrix for the graph.



Nodes	Branches					
	1	2	3	4	5	6
1	1	1	0	0	0	0
2	0	-1	1	1	0	0
3	0	0	0	-1	0	1
4	-1	0	-1	0	1	0
5	0	0	0	0	-1	-1

Complete Incidence Matrix

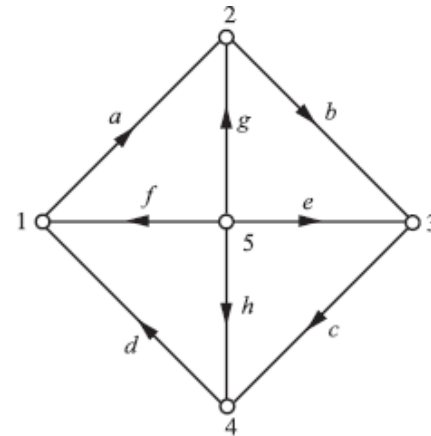
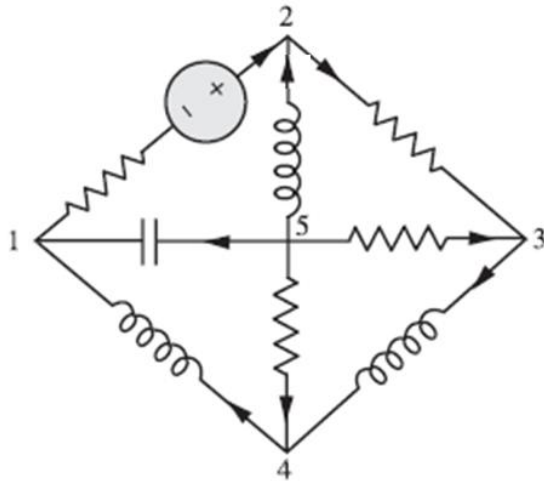
Reduced Incidence Matrix

$$[A] = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$$

$$[A_R] = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ -1 & 0 & -1 & 0 & 1 & 0 \end{bmatrix}$$

Network Graphs

6. Write the complete incidence matrix for the graph.



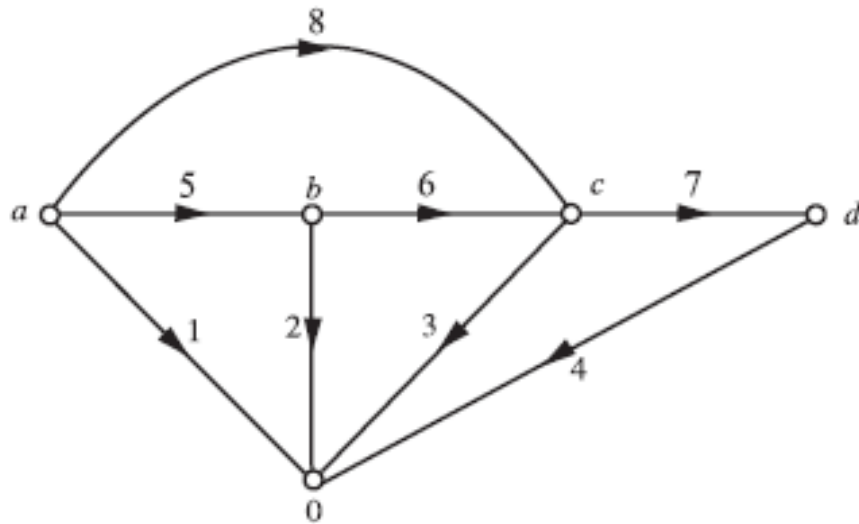
Nodes	Branch numbers							
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>
1	+1	0	0	-1	0	-1	0	0
2	-1	+1	0	0	0	0	-1	0
3	0	-1	+1	0	-1	0	0	0
4	0	0	-1	+1	0	0	0	-1
5	0	0	0	0	+1	+1	+1	+1

$$[A] = \begin{bmatrix} +1 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \\ -1 & +1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & +1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & +1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & +1 & +1 & +1 & +1 \end{bmatrix}$$

Network Graphs

7. For the incidence matrix given, draw the graph.

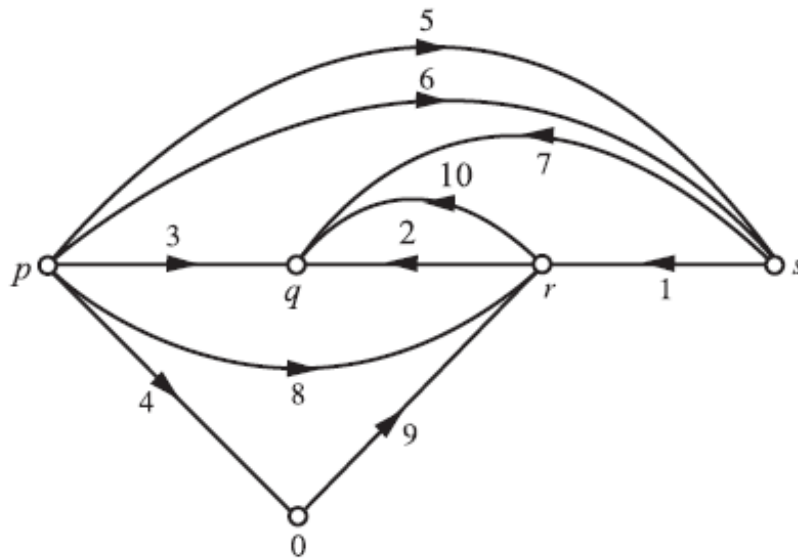
$$[A] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 \end{bmatrix} \end{matrix}$$



Network Graphs

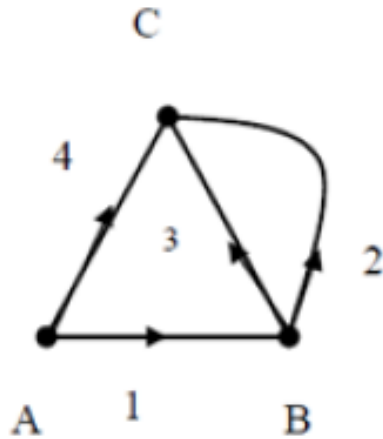
8. For the incidence matrix given, draw the graph.

$$[A] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{matrix} \\ \begin{matrix} p \\ q \\ r \\ s \end{matrix} & \left[\begin{array}{cccccccccc} 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 \\ 1 & 0 & 0 & 0 & -1 & -1 & 1 & 0 & 0 & 0 \end{array} \right] \end{matrix}$$



Network Graphs

9. Determine the number of trees in the given network



Node No.	Branches			
	1	2	3	4
A	1	0	0	1
B	-1	1	1	0
C	0	-1	-1	-1

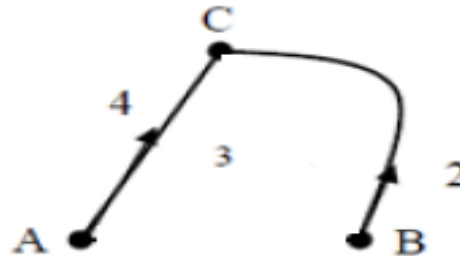
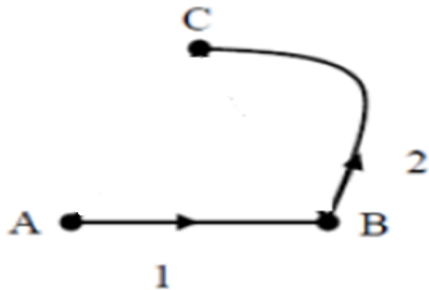
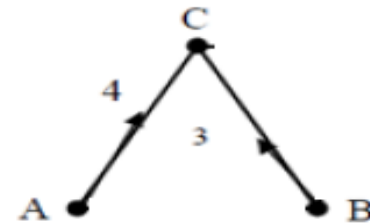
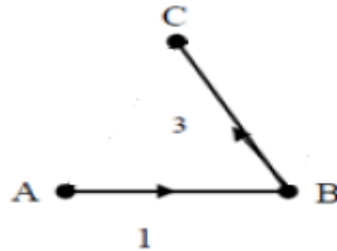
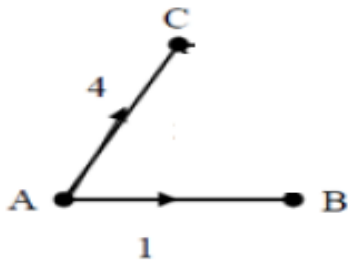
$$[A] = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 \\ 0 & -1 & -1 & -1 \end{pmatrix}$$

Network Graphs

$$[A_R] = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 \end{pmatrix}$$

$$\mathbf{A}_R \mathbf{A}_R^T = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1+1 & -1 \\ -1 & 1+1+1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \begin{vmatrix} 2 & -1 \\ -1 & 3 \end{vmatrix} = (6-1) = 5$$

Number of possible trees are 5.

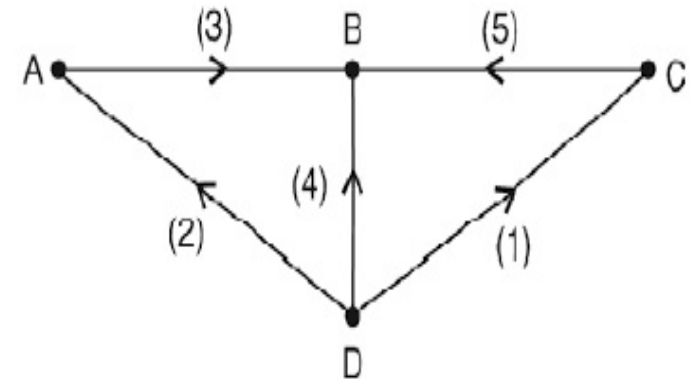


Network Graphs

10. For the given matrix, draw the oriented graph. How many trees are possible for this graph?

$$[A_R] = \begin{bmatrix} 0 & -1 & +1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & +1 \end{bmatrix}$$

$$[A] = \begin{array}{ccccc} \begin{bmatrix} 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & +1 \\ +1 & +1 & 0 & +1 & 0 \end{bmatrix} & \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{matrix} \text{Nodes} \downarrow \\ \end{matrix} \\ \begin{matrix} (1) & (2) & (3) & (4) & (5) \end{matrix} & \leftarrow \text{Branches} & \end{array}$$



Network Graphs

$$\begin{aligned}\text{No. of trees} &= \det \left\{ \begin{bmatrix} 0 & -1 & +1 & 0 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & +1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ +1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & -1 & +1 \end{bmatrix} \right\} \\ &= \det \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix} \\ &= 2(6 - 1) + 1(-2 - 0) + 0(1 - 0) \\ &= 10 - 2 = \mathbf{8}.\end{aligned}$$

Next class: Tie-set matrix